

Abstract

- A centralized platform that digitizes hostel and mess operations for students, wardens and admins.
- A microservices backend with polyglot databases reduces manual work, billing errors and food wastage.

Problem Statement & Objectives

- Problem:** Manual registers and spreadsheets cause data redundancy, inaccurate mess bills and food wastage.
- Objective 1:** Centralize hostel, room and student records.
- Objective 2:** Automate attendance, headcounts and billing.
- Objective 3:** Improve student-warden communication.
- Scope:** Students, wardens, admins; rooms, mess, complaints, leaves, payments.

Existing vs Proposed System

Feature	Existing System	Proposed System
Records	Paper registers, spreadsheets	Centralized database
Attendance	Manual, hard to track	Automated tracking
Mess billing	Error-prone calculation	Accurate, automated
Headcount	Inaccurate; food wasted	Real-time mess counts
Communication	Poor student-warden link	Complaints, leaves, alerts

Feasibility Study

- Technical:** Mature, well-documented tools (FastAPI, Node.js, Spring Boot, PostgreSQL).
- Economic:** Fully open-source stack with free-tier hosting and free CI/CD via GitHub Actions.
- Operational:** Simple role-based dashboards for each user type; containerized services are easy to maintain.
- Schedule:** Phased delivery – design, services, dashboards, testing – within the trimester timeline.

Results & Testing

- APIs tested with Thunder Client
- CRUD, JWT login and role-based access validated
- Input validation and error handling

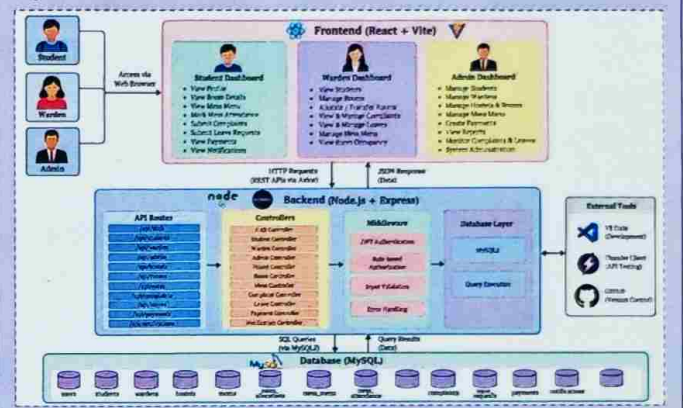
Conclusion

- Centralized hostel & mess platform that automates attendance, billing and headcounts.
- Microservices and polyglot databases give a scalable, production-ready base.

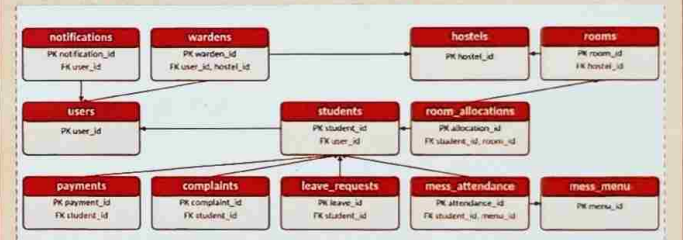
Tools & Technologies

- Front-End:** React (Vite) role-based dashboards
- Back-End:** FastAPI gateway with Node.js/Express and Spring Boot services; JWT auth
- Database:** PostgreSQL / MySQL (relational)
- Tools:** Git & GitHub, VS Code, Thunder Client

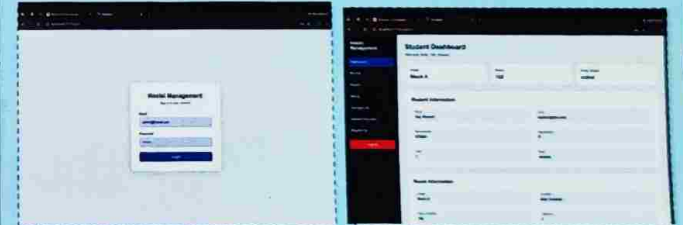
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Deploy on Kubernetes with CI/CD
- Mobile app with push notifications
- Analytics on occupancy and mess usage